

Tianshu Kuai

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Education

Sep 2024 – Current	PhD in Computer Science , Université de Montréal & Mila <i>GPA: 4.30 / 4.30</i> <i>Advisor: Prof. Noam Aigerman</i>
Sep 2022 – Dec 2023	MSc in Applied Computing , University of Toronto <i>GPA: 4.00 / 4.00</i> <i>Advisor: Prof. Igor Gilitschenski</i>
Sep 2017 – Apr 2022	Bachelor of Applied Science in Engineering Science , University of Toronto Robotics Major , Artificial Intelligence Minor <i>Major GPA: 3.80 / 4.00, cGPA: 3.73 / 4.00</i> <i>Thesis Advisor: Prof. Steven L. Waslander</i>

Publications

2026	[5] T. Kuai , A. Maesumi, D. Ritchie, and N. Aigerman, “Matérn Noise for Triangulation-Agnostic Flow Matching on Meshes,” <i>ACM Transactions on Graphics (SIGGRAPH)</i> 2026.
2025	[4] T. Kuai , S. Honari, I. Gilitschenski, and A. Levinshtein, “Towards Unsupervised Blind Face Restoration using Diffusion Prior,” <i>WACV</i> 2025.
2023	[3] T. Kuai , A. Karthikeyan, Y. Kant, A. Mirzaei, and I. Gilitschenski, “CAMM: Building Category-Agnostic and Animatable 3D Models from Monocular Videos,” <i>CVPRW</i> 2023. [2] A. Mahmoud, J. S. K. Hu, T. Kuai , A. Harakeh, L. Paull, and S. L. Waslander, “Self-Supervised Image-to-Point Distillation via Semantically Tolerant Contrastive Loss,” <i>CVPR</i> 2023.
2022	[1] J. S. K. Hu, T. Kuai , and S. L. Waslander, “Point Density-Aware Voxels for LiDAR 3D Object Detection,” <i>CVPR</i> 2022.

Experience

May 2023 - Apr 2024	Samsung AI Center Toronto Research Intern <i>Supervised by Dr. Alex Levinshtein, Samsung AI Center Toronto</i> Research on diffusion model based real-world image restoration and enhancement [4]
May 2022 - Apr 2024	University of Toronto 3D Computer Vision Researcher <i>Supervised by Prof. Igor Gilitschenski, Toronto Intelligent Systems Lab (TISL)</i> - Research on 3D scene representation and manipulation - Proposed a template-free method [3] for building animatable 3D models for arbitrary types of articulated and deformable objects from a collection of monocular videos, which allows users to animate reconstructed objects in 3D for content creation

May 2021 - Apr 2023	University of Toronto Computer Vision Researcher <i>Supervised by Prof. Steven L. Waslander, Toronto Robotics and Artificial Intelligence Lab (TRAILab)</i> - Research on self-supervised LiDAR semantic segmentation for autonomous driving, and contributed to the development of a novel method [2] that outperforms state-of-the-art 2D-to-3D representation learning frameworks - Designed and supported the development of high-performance LiDAR 3D object detection models for autonomous vehicles. PDV [1] achieved state-of-the-art multi-class 3D object detection results on Waymo Open Dataset upon publication
July 2021 - June 2022	aUToronto Computer Vision Engineer <i>University of Toronto Autonomous Driving Group, SAE/GM AutoDrive Challenge</i> - Research on fast and lightweight 3D perception models on collected data - Worked on deploying real-time perception models on autonomous vehicles
May 2020 - May 2021	Qualcomm Machine Learning Research Intern <i>Supervised by Dr. Shaojie Zhuo, Machine Learning Research Team</i> - Proposed several efficient deep learning models for audio processing - Applied state-of-the-art methods for neural network compression - Contributed to NPU software compiler pipeline development

Academic Service

Conference Reviewer | CVPR, ECCV, WACV, AAAI

Honors

Jan 2020	University of Toronto Engineering Competition Awarded the second prize in the senior design competition
Mar 2019	NSERC Undergraduate Student Research Award Undergraduate student research award from Natural Sciences and Engineering Research Council of Canada (NSERC)
Feb 2019	University of Toronto Excellence Award Awarded to University of Toronto undergraduate students based on research aptitude
Sep 2017	University of Toronto Engineering Entrance Scholarship Scholarship for top engineering candidates pursuing studies at the University of Toronto